McMaster University

ENGINEERING
McMaster-Mohawk
Bachelor of Technology
Partnership

SMRTTECH 4ES3 – Embedded Systems

Extra Lab: Introduction to **Electronic Design Automation** (EDA)

A. Objective

In this lab, you will learn how to create a complete electronic design workflow using KiCad, an Electronic Design Automation (EDA) software suite. You will practice building a schematic, assigning footprints, generating a PCB layout, running electrical rule checks, and exporting manufacturing files. By the end of the lab, you will understand the essential steps required to design, verify, and prepare a simple printed circuit board (PCB) for fabrication.

B. Materials

This lab requires the use of:

- Laptop or Computer
- KiCad 9.0 software
- Lab file provided

C. Installing KiCad

a. Download software from KiCad official website.
   https://www.kicad.org/download/

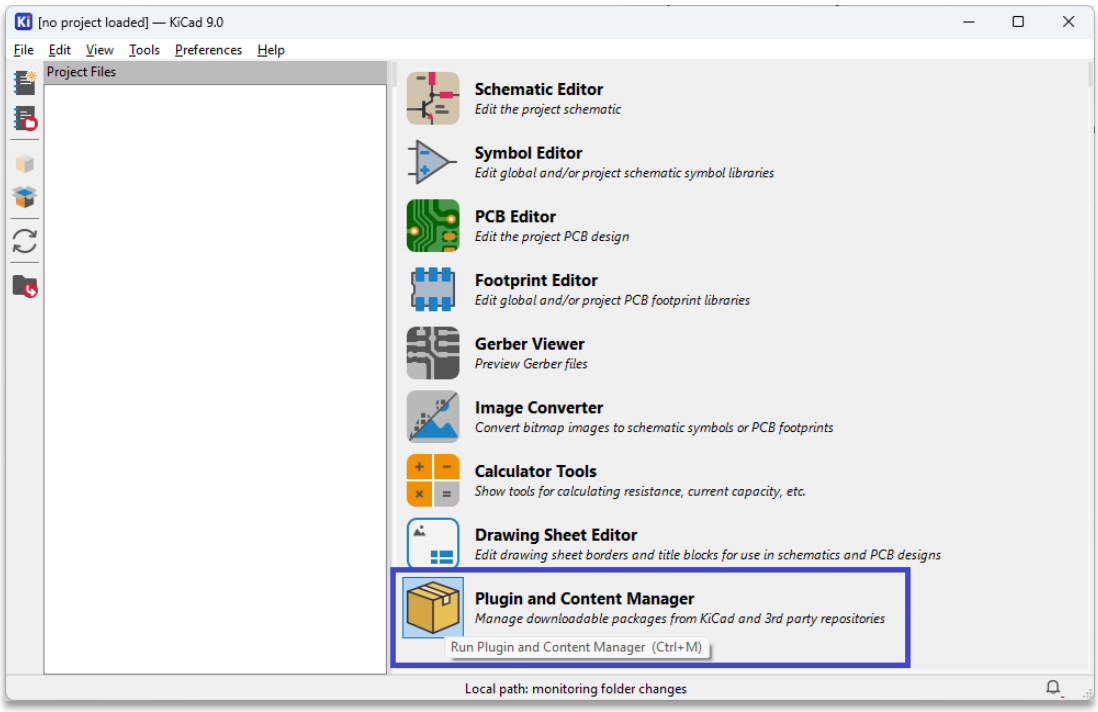
b. Select your operating system

c. Select your regional download
   i. Select the North America Git Hub download

d. After the download is complete, open the file
   i. Follow the setup instructions. Click **Next**.
   ii. Selected **For me**. Click **Next**.
   iii. Ensure all components are selected. Click **Next**.
   iv. Do not change the destination folder. Click **Install**.
   v. Wait for Installation to complete. Click **Finish**.

e. Install the Necessary plugin

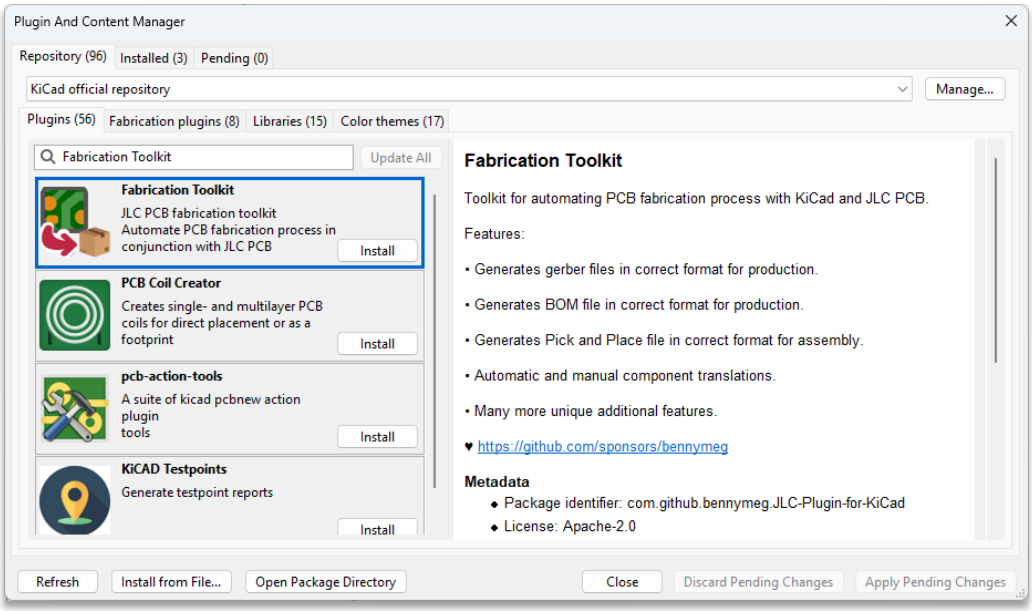
i. Click **Plugin and Content Manager.**



ii. In the search bar, type **Fabrication Toolkit**

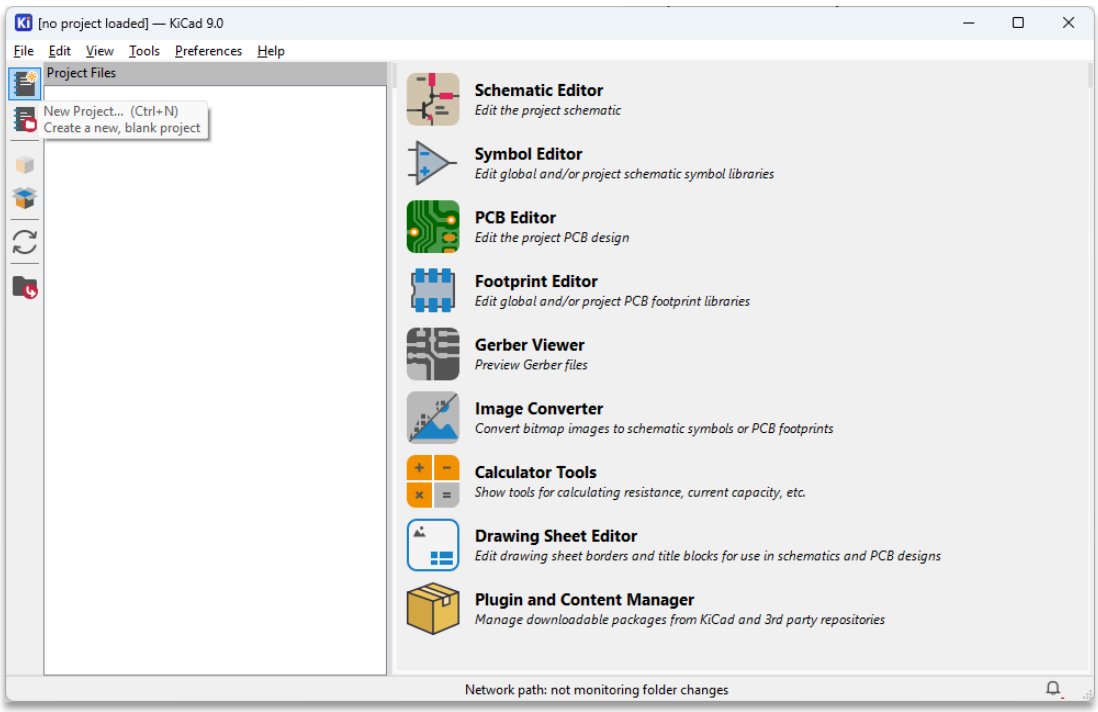
iii. Click **Install** and **Apply Pending Changes**

iv. Close out of then Plugin Menu

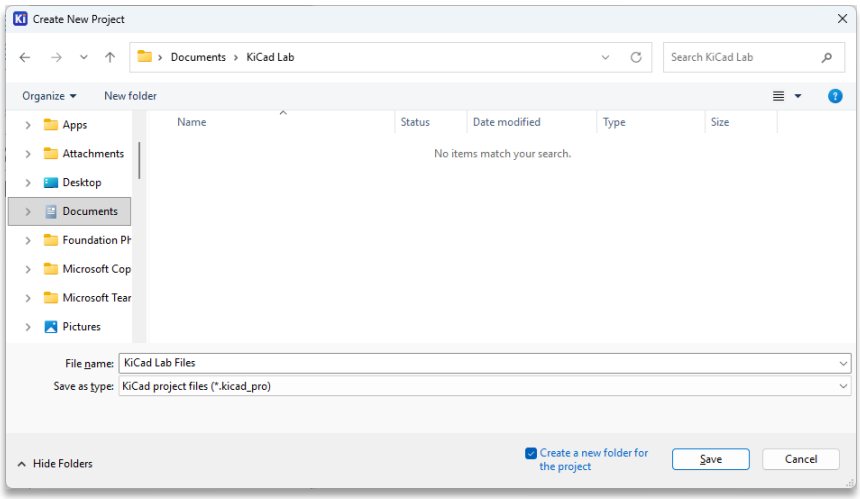


D. Creating Project Files

a. Start by creating a project folder
   i. Under “File” you can create a new project.

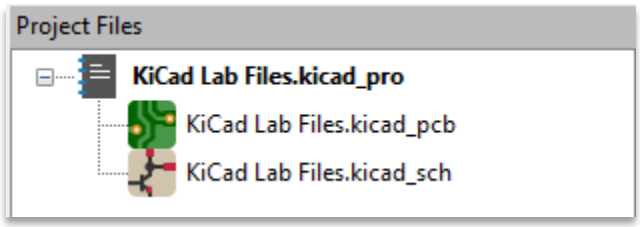


b. Find a location for the project folder.
   i. In this example, we used the Documents folder.
   ii. Name the project.
   iii. Click Save



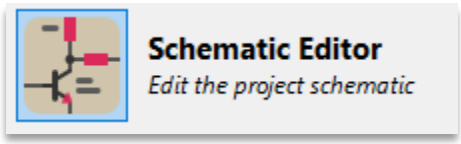
c. This will create a few files in the Project File viewer

i. The main file being the project file itself
ii. Followed by the PCB file
iii. Last is the Schematic file

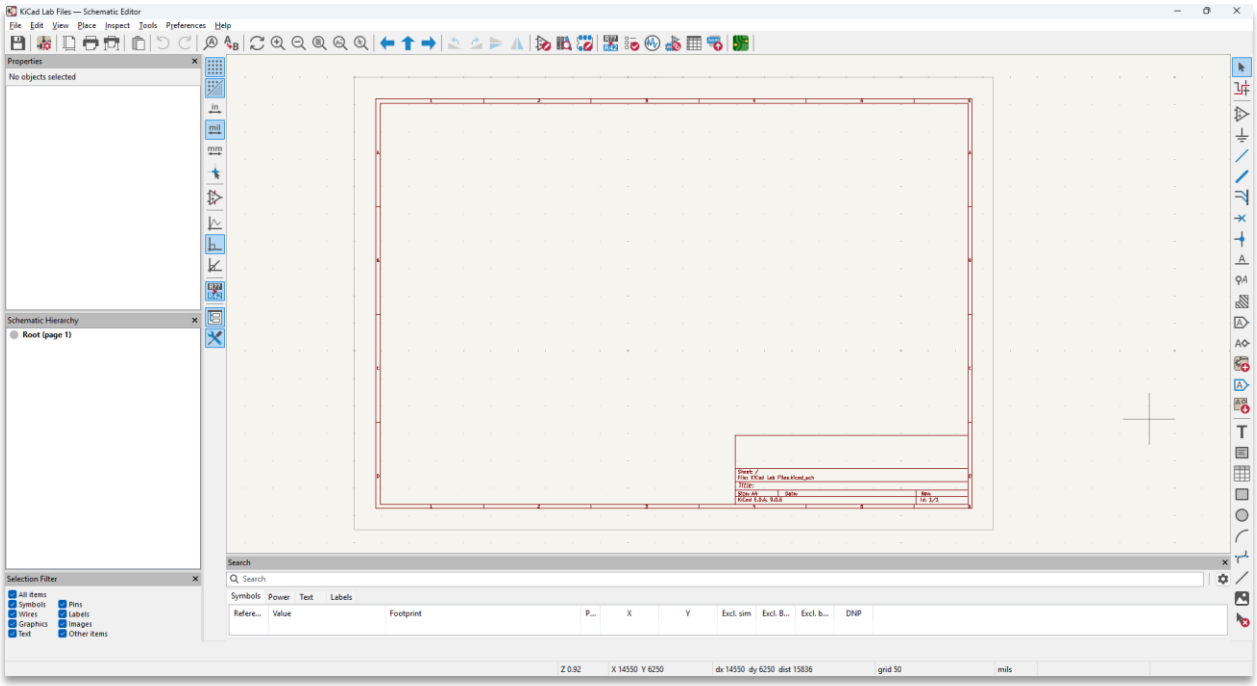


E. Creating your schematic

a. Open the Schematic Editor

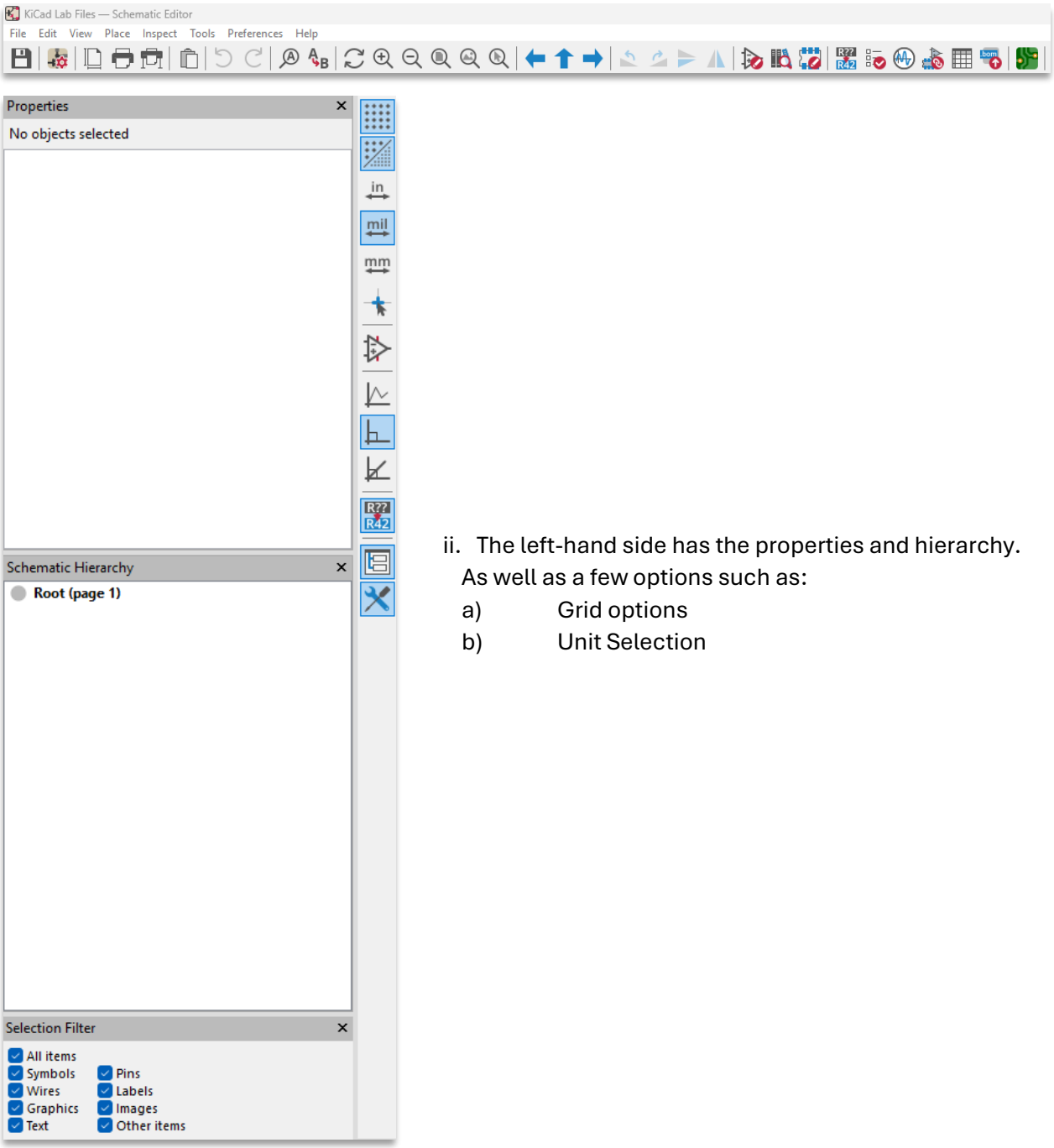


i. This will open your electrical schematic drawings
ii. Although blank right now, this is where you will add your components in their simplest form, Symbols



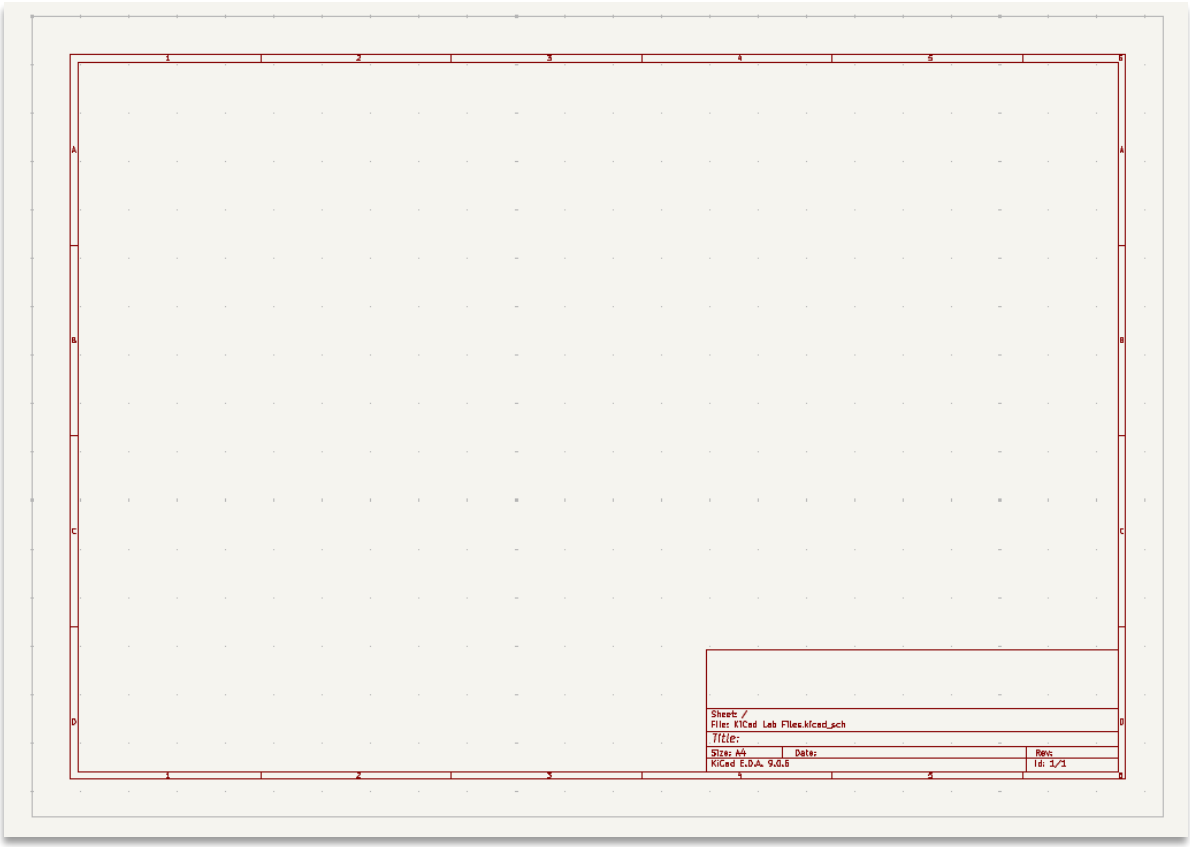
b. This screen may look busy, Let's break it down
   i. The top bar has all your basic options such as:

a) Saving your file
b) Schematics properties
c) Printing
d) Moving and zooming options
e) Symbols and rule checkers



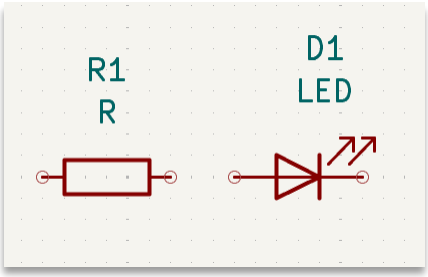
ii. The left-hand side has the properties and hierarchy. As well as a few options such as:
a) Grid options
b) Unit Selection

iii. The main window is where we will place the Symbols to construct the schematic
   a) Double clicking on the title block will allow us to make changes



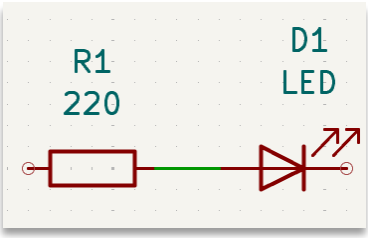
iv. Lastly, the right-hand side will be the most important to understand. With the main feature to work with, such as:
   a) Symbol Placement tool
   b) Power Symbol tools
   c) Wire and Bus tools
   d) Labeling tool
   e) Drawing and Text tools

c. Add a few Symbols
   i. Press **A**, or *Place Symbol* on the right side
   ii. This will open the library menu
   iii. Add a resistor and an LED to the schematic

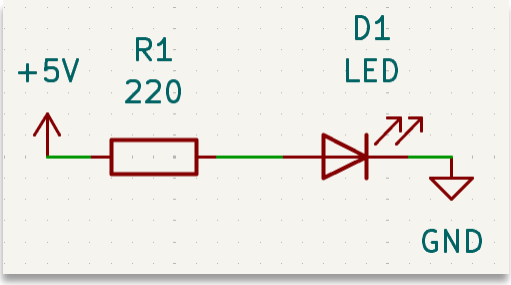


   iv. Clicking on a symbol will bring up the properties in the left side menu, or Press **E** while selected the component
   v. Change the value of the resistor to 220 Ohms

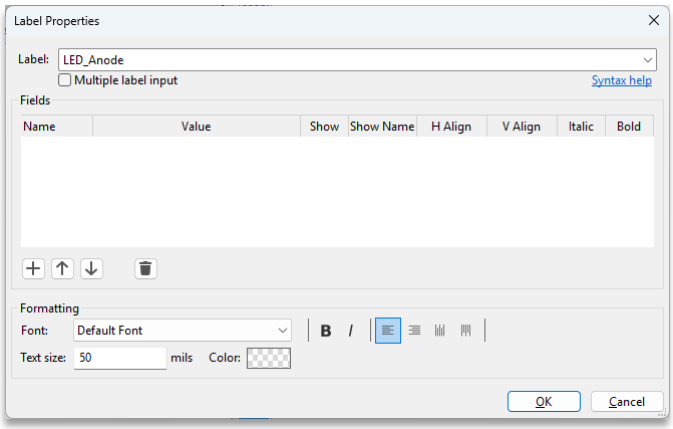
d. Connect the Symbols
   i. Using the Wire tool on the left side menu, Connect the resistor to the LED



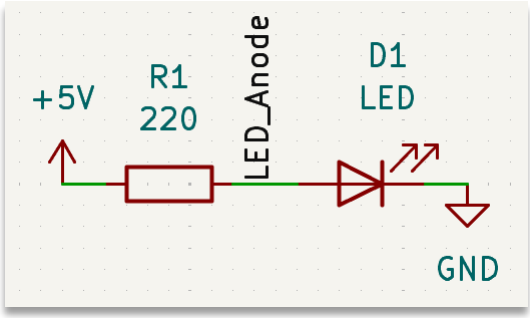
e. Add power Symbols
   i. The Resistor will need a Voltage input
   ii. The LED will need a Ground connection
   iii. Press **P**, or open *Place Power Symbol*
   iv. Search for **+5V**, then place that
   v. Search for **GND**, select one and place it
   vi. Use the Wire tool to connect them



f. Naming Wires
   i. The wire in between the Resistor and LED currently isn't named
   ii. Use either the *Place Net Labels,* or *Place Global Label*
   iii. Label the wire to your design criteria, in this case call it "LED_Anode"



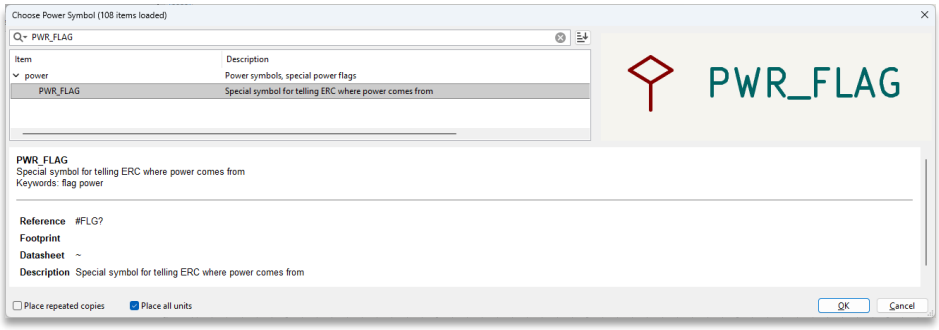
   iv. Press **OK**, then place the label on the wire. Use **R** to rotate the label

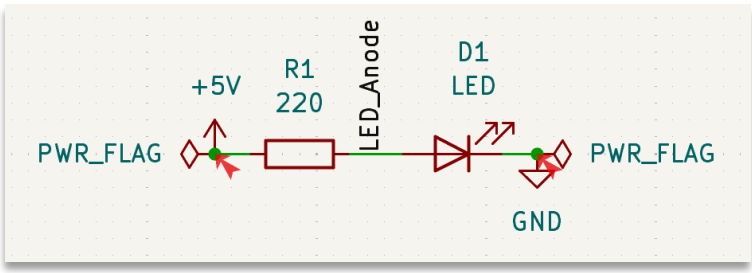


g. Electrical Rule Checker
   i. After creating the schematic, we can use built in tools to check that the schematic does not violate any rules and does not have errors between symbols
   ii. Open **Electrical Rule Checker** from the top menu
   iii. Press **Run ERC**
   iv. Do we have errors?
   v. Double Clicking on the error will take you to the wire.

h. PWR_FLAG
   i. A special feature of KiCad is the PWR_FLAG. This flag allows us to tell the Electrical Rule Checker where the power is coming from, **Vcc,** and where the power is going, **GND.**

ii. Add two PWR_FLAG by opening the Power Symbol Menu and searching





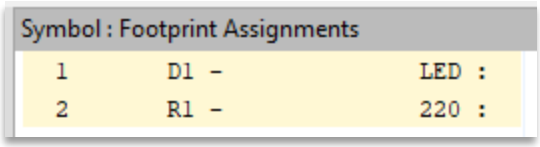
iii. Rerun the Electrical Rules Checker. Are there errors?

i. Add Footprints

i. Some Symbols you place will have footprints preselected, as we selected generic components, we will need to assign footprint to them

ii. Open **Assign Footprints...** on the top menu

iii. The Middle Menu will be a list of all symbols you placed in the schematic



iv. Select the LED under D1, Using the top menu to search for LED footprints. Use either:
   a) **LED_THT**, for through hole LEDs
   b) **LED_SMD**, for surface mount LEDs

v. Select a footprint by double clicking an item in the right-hand menu

vi. Repeat steps for Resistor using either:
   a) **Resistor_THT**
   b) **Resistor_SMD**

vii. Select **OK**

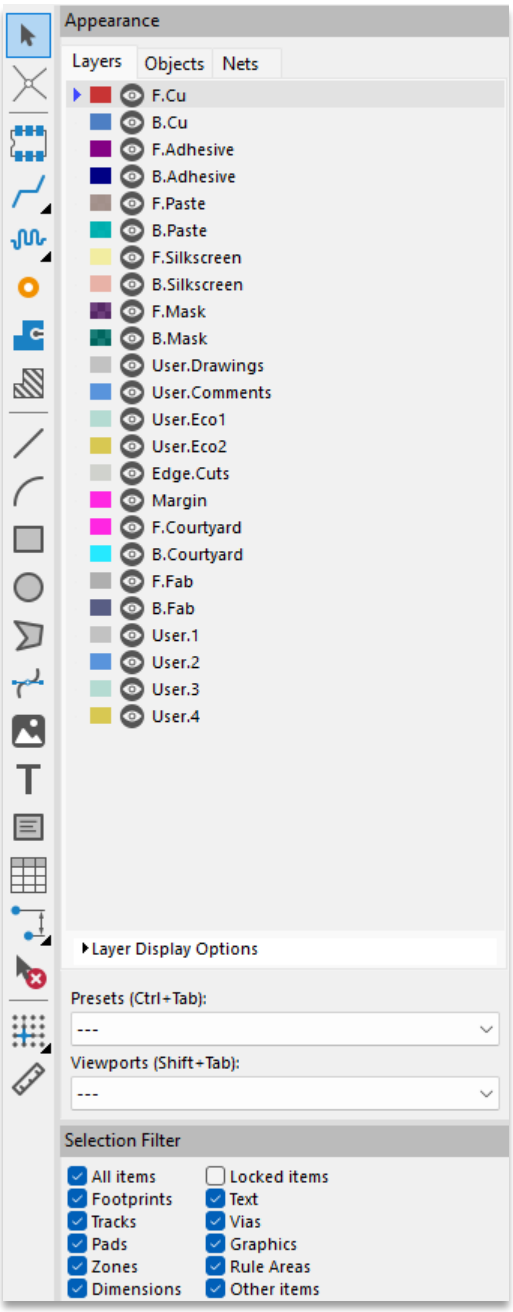
j. You have now completed a basic schematic! Use **Switch to PCB Editor**

F. Create your PCB

The PCB Editor uses a similar menu system to the schematic editor.

The one notable exception is the right-hand side menu has the Appearance table.

This is where we want to select the layers of the PCB.



a. Start off by selecting **Edge Cuts.**

b. Using the **Draw Rectangle** tool, Make a square on the PCB editor of at least 10mm by 10mm.

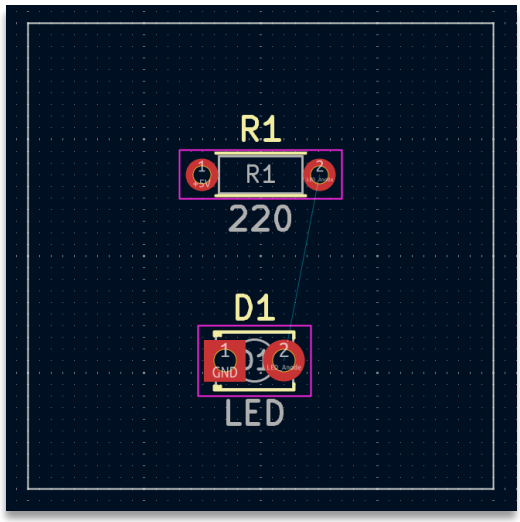
   This will create the outside edges of the PCB

c. Either Press **F8**, or Click **Update PCB from Schematic** in the top menu.

d. This menu will show you the change you made in the Schematic editor and will update the PCB with the footprint selected.

e. Click **Update PCB.**

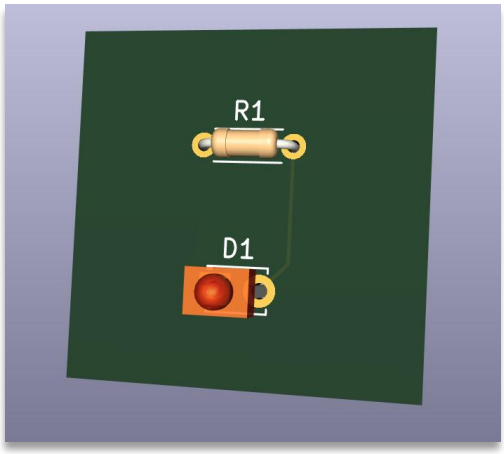
f. This will add the Footprints of your symbols to the PCB. Place them by clicking anywhere on the PCB



g. The blue line, in the PCB layout, shows shortest path between points that need to be connected
   i. Press **X** or use the **Route Single Track** on the right-hand menu
   ii. Click on one of the two points and connect them.

h. Like the Electrical Rule Checker, The PCB Editor has the **Design Rule Checker** in the top menu

i. To view the PCB in 3D, Press **3D Viewer** in the top menu
   i. Take a screenshot and submit it in the lab report



j. To create Gerber files of the design, select **Fabrication Toolkit**
   i. Press **Generate**
   ii. Submit the **.Zip** file